

Problem 1, Cell 1: the planar case $d = 2$

Official statement. A line here always means a line through the origin of $\mathbb{R}^d$. The angle between two lines ℓ, ℓ' is the acute (non-obtuse) angle $\theta(\ell, \ell') \in [0, \pi/2]$ between them. If ℓ, ℓ' are spanned by unit vectors x, x' , then $\theta(\ell, \ell') = \arccos |\langle x, x' \rangle|$. For lines $\ell_1, \dots, \ell_N$ in $\mathbb{R}^d$ (repetitions allowed), write $S(\ell_1, \dots, \ell_N) = \sum_{1 \leq i < j \leq N} \theta(\ell_i, \ell_j)$.

We begin in the plane ($d = 2$), where the conjectured optimum splits the N lines as evenly as possible between two perpendicular directions. Let $\ell_1, \dots, \ell_N$ be lines in $\mathbb{R}^2$. Prove that $S(\ell_1, \dots, \ell_N) \leq (\pi/2) \cdot \lfloor N^2/4 \rfloor$.

Theorem 1 below is exactly this statement. It is proved for arbitrary lines in $\mathbb{R}^2$, and lines through the origin are a special case. For lines spanned by unit vectors x, x' , the angle $\angle$ used below equals $\arccos |\langle x, x' \rangle| = \theta$. Repeated lines are allowed.

Known result. The planar case is known [2, 1]; the proof below is self-contained.

Setting. Let $\ell_1, \dots, \ell_N$ be lines in $\mathbb{R}^2$. The angle $\angle(\ell_i, \ell_j) \in [0, \pi/2]$ between two lines is the smaller of the two angles formed by their directions (it is 0 for parallel lines, and it depends only on the directions of the lines). Put

$$S = \sum_{1 \leq i < j \leq N} \angle(\ell_i, \ell_j).$$

Theorem 1. $S \leq \frac{\pi}{2} \left\lfloor \frac{N^2}{4} \right\rfloor$.

1 Directions as points of a circle of length π

Let $\mathbb{T} = \mathbb{R}/\pi\mathbb{Z}$, with the Lebesgue measure λ inherited from $[0, \pi)$ (so $\lambda(\mathbb{T}) = \pi$) and the quotient map $q: \mathbb{R} \rightarrow \mathbb{T}$. The direction of a line is determined by an angle $\varphi \in \mathbb{R}$ with the x -axis, up to adding multiples of π ; hence each ℓ_i has a well-defined direction $\varphi_i \in \mathbb{T}$.

For $u \in \mathbb{T}$ define $\|u\| = \min\{|t| : t \in \mathbb{R}, q(t) = u\} \in [0, \pi/2]$. If $t \in [0, \pi)$ is the representative of u , then $\|u\| = \min(t, \pi - t)$. By definition of the angle between two lines,

$$\angle(\ell_i, \ell_j) = \|\varphi_i - \varphi_j\|. \tag{1}$$

Remark 2. Any half-open interval $[s, s + \pi) \subset \mathbb{R}$ is mapped by q bijectively onto $\mathbb{T}$, and q restricted to it preserves the measure of Borel sets. For $s = 0$ this is the definition of λ , and for general s it follows by translation invariance of the Lebesgue measure. This is the only fact about “wrap-around” we shall need.

2 The arcs

For each i choose a real representative p_i of φ_i and set

$$I_i = q\left([p_i - \frac{\pi}{4}, p_i + \frac{\pi}{4})\right) \subset \mathbb{T}.$$

This set does not depend on the choice of p_i (changing p_i by $k\pi$ translates the interval by $k\pi$). By Remark 2, $\lambda(I_i) = \pi/2$, since $[p_i - \frac{\pi}{4}, p_i + \frac{\pi}{4})$ is contained in a half-open interval of length π on which q is injective and measure preserving. (This value is not used below; it only shows that each arc covers exactly half of $\mathbb{T}$.) We write $\mathbf{1}_i$ for the indicator function of I_i .

Lemma 3 (Symmetric difference). *For all $i \neq j$,*

$$\lambda(I_i \triangle I_j) = \int_{\mathbb{T}} |\mathbf{1}_i(x) - \mathbf{1}_j(x)| dx = 2 \angle(\ell_i, \ell_j).$$

Proof. The first equality holds because $|\mathbf{1}_i - \mathbf{1}_j|$ is the indicator function of $I_i \triangle I_j$. For the second, let $\theta = \angle(\ell_i, \ell_j) = \|\varphi_j - \varphi_i\| \in [0, \pi/2]$. Let $t \in [0, \pi)$ be the representative of $\varphi_j - \varphi_i$. Then $\theta = t$ or $\theta = \pi - t$. In the second case, $\varphi_i - \varphi_j$ has representative $\pi - t = \theta$ (if $t > 0$; if $t = 0$ the first case holds). Since the statement is symmetric in i and j , we may swap the two indices if needed and assume

$$\varphi_j - \varphi_i = q(\theta), \quad 0 \leq \theta \leq \pi/2.$$

Choose the representatives p_i and $p_j = p_i + \theta$. Let $J_i = [p_i - \frac{\pi}{4}, p_i + \frac{\pi}{4})$ and $J_j = [p_i + \theta - \frac{\pi}{4}, p_i + \theta + \frac{\pi}{4})$, so that $I_i = q(J_i)$ and $I_j = q(J_j)$.

No wrap-around. Because $0 \leq \theta \leq \pi/2$, both J_i and J_j lie in the half-open interval $F = [p_i - \frac{\pi}{4}, p_i + \frac{3\pi}{4})$, which has length exactly π . (Here we use $\theta + \frac{\pi}{4} \leq \frac{3\pi}{4}$.) By Remark 2, $q|_F$ is a measure-preserving bijection $F \rightarrow \mathbb{T}$. Being a bijection, it commutes with set operations: $q(J_i) \triangle q(J_j) = q(J_i \triangle J_j)$. Hence $\lambda(I_i \triangle I_j)$ equals the Lebesgue measure of $J_i \triangle J_j$ computed in $\mathbb{R}$.

Computation in $\mathbb{R}$. Since $\theta \leq \pi/2$, the left end of J_j satisfies $p_i + \theta - \frac{\pi}{4} \leq p_i + \frac{\pi}{4}$, so

$$J_i \setminus J_j = [p_i - \frac{\pi}{4}, p_i + \theta - \frac{\pi}{4}), \quad J_j \setminus J_i = [p_i + \frac{\pi}{4}, p_i + \theta + \frac{\pi}{4}).$$

Each has length θ . (If $\theta = \pi/2$ the two intervals are disjoint and adjacent, and the formulas still hold. If $\theta = 0$ both sets are empty.) Therefore $\lambda(I_i \triangle I_j) = 2\theta$. $\square$

3 Counting pairs

Define the covering function

$$A(x) = \#\{i \in \{1, \dots, N\} : x \in I_i\} = \sum_{i=1}^N \mathbf{1}_i(x), \quad x \in \mathbb{T}.$$

It is a finite sum of indicator functions of arcs, so it is a bounded measurable (step) function with integer values in $\{0, 1, \dots, N\}$.

Lemma 4 (Pair count). *For every $x \in \mathbb{T}$,*

$$\sum_{1 \leq i < j \leq N} |\mathbf{1}_i(x) - \mathbf{1}_j(x)| = A(x) (N - A(x)).$$

Proof. Fix x . The number $|\mathbf{1}_i(x) - \mathbf{1}_j(x)|$ is 1 if exactly one of I_i, I_j contains x , and 0 otherwise. So the left-hand side counts unordered pairs $\{i, j\}$ with exactly one of the arcs containing x . Let $P = \{i : x \in I_i\}$, so $|P| = A(x)$. Such a pair consists of one index in P and one index outside P . Each unordered pair of this kind corresponds to exactly one element of $P \times (\{1, \dots, N\} \setminus P)$, and conversely. Hence there are $A(x)(N - A(x))$ such pairs. $\square$

Lemma 5. *For every integer a with $0 \leq a \leq N$ we have $a(N - a) \leq \lfloor N^2/4 \rfloor$.*

Proof. $a(N - a) = \frac{N^2}{4} - (a - \frac{N}{2})^2$. If N is even this is at most $N^2/4 = \lfloor N^2/4 \rfloor$. If N is odd, $a - \frac{N}{2}$ is a half-integer, so $(a - \frac{N}{2})^2 \geq \frac{1}{4}$ and $a(N - a) \leq \frac{N^2 - 1}{4} = \lfloor N^2/4 \rfloor$. $\square$

4 Proof of Theorem 1

By (1), Lemma 3, linearity of the integral (a finite sum), and Lemma 4,

$$2S = \sum_{i < j} 2\angle(\ell_i, \ell_j) = \sum_{i < j} \int_{\mathbb{T}} |\mathbf{1}_i - \mathbf{1}_j| dx = \int_{\mathbb{T}} \sum_{i < j} |\mathbf{1}_i - \mathbf{1}_j| dx = \int_{\mathbb{T}} A(x)(N - A(x)) dx.$$

By Lemma 5 the integrand is at most $\lfloor N^2/4 \rfloor$ at every point, and $\lambda(\mathbb{T}) = \pi$. Hence $2S \leq \pi \lfloor N^2/4 \rfloor$, that is, $S \leq \frac{\pi}{2} \lfloor N^2/4 \rfloor$. $\square$

Remark 6 (Sharpness). Take $\lfloor N/2 \rfloor$ lines parallel to the x -axis and $\lceil N/2 \rceil$ lines parallel to the y -axis. Exactly $\lfloor N/2 \rfloor \lceil N/2 \rceil = \lfloor N^2/4 \rfloor$ pairs are perpendicular, and all other pairs have angle 0. So equality holds and the bound is best possible.

References

- [1] D. Bilyk and R. W. Matzke, *On the Fejes Tóth problem about the sum of angles between lines*, Proc. Amer. Math. Soc. **147** (2019); arXiv:1801.07837.
- [2] F. Fodor, V. Víggh and T. Zarnócz, *On the angle sum of lines*, Arch. Math. **106** (2016), 91–100.